CONTAMINATED-SITE HAZARD

Superfund (NPL) Sites

A Superfund listing identifies a serious cleanup program, but the site boundary is not the same as the contaminant plume or exposure area.

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1 Understand the factor

What it is - and why it matters

Superfund is the federal program for investigating and cleaning some of the nation's most contaminated sites under the Comprehensive Environmental Response, Compensation, and Liability Act. The National Priorities List, or NPL, contains sites eligible for long-term federal remedial action. A site may include former factories, landfills, mines, military facilities, waterways, or groundwater plumes. Risk depends on the contaminants, media affected, cleanup stage, institutional controls, and whether people can contact contaminated soil, dust, vapor, sediment, food, or water.

How risk can reach a household

- Contaminated groundwater can migrate beyond the facility boundary and affect drinking-water wells or create vapor-intrusion concerns.
- Polluted soil or sediment can be contacted directly, tracked indoors as dust, eroded, or disturbed during gardening and construction.
- Volatile chemicals can move through soil gas into buildings when subsurface conditions and foundations create a pathway.
- Flooding, wildfire, excavation, or failing caps and drainage systems can redistribute contaminated material.

What people often miss

- The drawn NPL polygon may represent the administrative site, not the full plume, study area, operable units, or off-site properties.
- Listing does not mean every nearby property is exposed, and cleanup completion does not always mean unrestricted use.
- Institutional controls may prohibit wells, excavation, residential use, gardening, or soil disturbance and can run with the land.
- A site's risk can decline substantially after cleanup, but monitoring and five-year reviews may continue for decades.

? Five checks before buying

1. Read the EPA site profile, Record of Decision, cleanup maps, five-year reviews, community updates, and institutional-control documents.
2. Identify specific contaminants, affected media, plume direction and depth, exposure assumptions, and the home's position relative to groundwater flow.
3. Search deed records and local planning files for land-use covenants, well restrictions, vapor systems, soil-management plans, or special permits.
4. Use an environmental professional to determine whether Phase I review, soil, groundwater, dust, or sub-slab vapor testing is appropriate.
5. Ask lenders and insurers how known contamination, remediation systems, disclosure, pollution exclusions, and resale may be treated.

IMPORTANT DISTINCTION Map proximity is a screening signal - not proof of exposure, damage, or insurability. Confirm the exact feature, current condition, pathway, and property-specific controls.

HEALTH, PROPERTY AND COVERAGE

How this factor can affect a household

FACTOR 10

H Health impacts

- Health effects vary widely: lead can affect neurological development; solvents may affect the nervous system and organs; some contaminants increase cancer risk.
- Children may have higher soil and dust contact, and private-well users can have a direct pathway if a plume reaches their water source.
- The presence of a chemical is not the same as exposure; concentration, route, frequency, duration, and vulnerability determine risk.
- Uncertainty, odors, cleanup activity, and fear about children's health can create persistent stress even when exposures are controlled.

P Property impacts

- Contamination can require caps, vapor systems, soil removal, well abandonment, water treatment, access agreements, and construction controls.
- Market stigma, disclosure, lender concerns, monitoring equipment, and land-use restrictions may affect value and redevelopment.
- Excavation may be slower and more expensive because soil must be characterized, handled, transported, and documented.
- Liability can be complex; residential-owner protections exist in some circumstances, but legal review may be prudent for impacted parcels.

I Insurance implications

- Standard homeowners insurance generally does not pay to clean pre-existing pollution or gradual contamination.
- Environmental impairment policies are specialized products and may exclude known conditions disclosed before policy inception.
- Sudden covered damage may be insured while the contaminant removal, land restoration, or government-ordered cleanup remains excluded or limited.
- The best protection is strong due diligence, clear agency status, professional testing where warranted, and documented controls.

Possible long-term health implications of buying nearby

- Long-term health impact depends on whether a complete exposure pathway remains after cleanup and controls, not merely the site's distance.
- Residents should understand what system must operate, who maintains it, what alarms mean, and how results are communicated.
- A well-characterized home with clean drinking water, controlled vapor, covered soil, and current monitoring can differ greatly from an untested property over a plume.
- Changes such as a basement remodel, new well, garden, excavation, flood, or utility trench can create pathways that did not previously exist.

Evidence lens

The toxicology of many Superfund contaminants is well established, but site-specific health interpretation requires current sampling and a complete source-pathway-receptor analysis.

Plan more carefully for

- Children
- Older adults
- Pregnancy
- Heart/lung conditions
- Mobility or power needs

Insurance reality: Availability, eligibility, exclusions, deductibles, and limits vary by state, carrier, property, and cause of loss. Get an address-specific written quote before the purchase contingency expires.

PRACTICAL RISK REDUCTION

Precautions and a real-life example

FACTOR 10

V Verify the actual risk

- Build a timeline from EPA decisions, cleanup construction, monitoring, five-year reviews, and any remaining restrictions.
- Confirm whether the parcel is inside a study area, plume, vapor-management zone, or institutional-control boundary.

P Protect the property

- Do not disturb capped soil, monitoring wells, vapor piping, or restricted areas without agency and professional approval.
- Maintain vapor systems, sump seals, drainage, and clean soil cover; keep operation manuals and maintenance records.

M Monitor and respond

- Review new sampling reports and agency notices; test private wells on the schedule recommended by regulators.
- Report odors, damaged caps, flooding, construction disturbance, system alarms, or unexplained indoor-air changes.

I Prepare insurance records

- Obtain written guidance on pollution and remediation exclusions and whether required mitigation equipment is covered after a sudden loss.
- Preserve environmental reports, disclosures, agency letters, legal documents, photographs, and maintenance logs.

REAL-LIFE EXAMPLE

Love Canal Environmental Emergency

1970s-1980s

Homes and a school had been built near a former chemical-disposal area in the Love Canal neighborhood of Niagara Falls, New York. Chemicals migrated from the disposal area, residents reported odors and health concerns, and federal actions ultimately included emergency declarations, relocation, containment, and long-term cleanup. The case helped drive creation of the federal Superfund program.

> **Why it matters:** Love Canal showed how historic disposal, incomplete land-use knowledge, groundwater and soil migration, and residential development can combine into a long-duration public-health and property crisis.

Open official incident record - epa.gov >

Questions to resolve before making a decision

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4. Use an environmental professional to determine whether Phase I review, soil, groundwater, dust, or sub-slab vapor testing is appropriate.

Official sources and mapping tools

- EPA: What is Superfund? - epa.gov >
- Search Superfund sites - epa.gov >
- EPA Superfund history - epa.gov >